

Titer Determination of $\frac{1}{2}$ H₂SO₄ 0.1 mol/L

Method for the standardization of 0.1 mol/L $\frac{1}{2}$ H₂SO₄ sulfuric acid using Tris(hydroxymethyl)aminomethane (TRIS, or also THAM) as the primary standard. The titration is monitored with a combined pH glass electrode.

Sample	Primary standard Tris(hydroxymethyl)aminomethane (TRIS/THAM) 50 – 100 mg
Compound	Tris(hydroxymethyl)aminomethane, TRIS/THAM, (HOCH ₂) ₃ CNH ₂ M = 121.14 g/mol, z = 1
Chemicals	98% H ₂ SO ₄ 50 mL Deionized water
Titrant	Sulfuric acid, H ₂ SO ₄ c($\frac{1}{2}$ H ₂ SO ₄) = 0.1 mol/L
Standard	--
Indication	DGi111-SC combined pH glass electrode
Chemistry	$\text{H}_2\text{SO}_4 + 2 (\text{HOCH}_2)_3\text{CNH}_2 \rightarrow 2 (\text{OHCH}_2)_3\text{CNH}_3^+ + \text{SO}_4^{2-}$
Calculation	$R = m/(\text{VEQ} \cdot c \cdot C)$ $C = M/(10 \cdot p \cdot z)$ VEQ= Titrant consumption (mL) c = Titrant concentration (mol/L) m = Sample mass (g) M = Sample molar mass (g/mol) p = Purity of the standard (%) z = Sample equivalent number
Waste disposal	Neutralization before final disposal as aqueous solution
Author, Version	Vineesh Pallath, MSG AnaChem, Version 2.0 Revised: C. De Caro, MSG AnaChem

Preparation and Procedures

CAUTION

- Use safety goggles, a lab coat and wear gloves.
- Ensure accurate cleaning of sensor is sufficient after each titration.

Preparation of 0.1 mol/L $\frac{1}{2}$ H₂SO₄:

- Add approx. 5 g 98% H₂SO₄ to a 1 L volumetric flask containing 500 mL deionized water.
- Heat will generate; cool to room temperature and fill up to 1000 mL with deionized water.

Preparation of sample solution:

- Weigh 0.05 – 0.10 g TRIS in to a glass titration beaker on the sample changer rack.
- Start the method.
- 50 mL of deionized water will be added by membrane pump of sample changer.
- The solution is stirred for 120 s and titration will proceed.
- After completion of titration, the electrode, stirrer and titration tubes will be rinsed by deionized water using the membrane pump.
- Subsequently, the electrode is conditioned for 30 s with deionized water in condition beaker.

Remarks

- The method parameters have been optimized for the sample of this application. It may be necessary to adapt the method to your specific sample.
- Glass beakers are recommended in order to avoid any interference during weighing due to electrostatic force.
- This method allows a fully automated analysis procedure. The method can be easily modified for manual operation. Select "Manual stand" in the method function "Titration stand".

Instruments

- Titration Excellence T50/T70/T90
(Other Titrators: With modifications or manual operations are necessary)
- Rondo 20 Sample Changer with PowerShower™ - (ME-51108003)
- XP205 Balance (ME-11106024)

Accessories

- 1 x 10mL DV1010 glass burette (ME-51107501)
- Glass titration beaker (ME- 00101446)
- LabX® pro titration software

Results

All results

Method-ID	TiterH2SO4
Sample	0.06485 g (1/6)
R1 (Titer)	0.99882
Sample	0.06239 g (2/6)
R1 (Titer)	0.99685
Sample	0.06850 g (3/6)
R1 (Titer)	1.00068
Sample	0.06323 g (4/6)
R1 (Titer)	1.00067
Sample	0.06714 g (5/6)
R1 (Titer)	0.99926
Sample	0.06284 g (6/6)
R1 (Titer)	0.99858

Statistics

Method-ID	VPTiterH2SO4
R1	Titer
Samples	6
Mean	0.99914
s	0.00144
srel	0.144 %

Titration curve

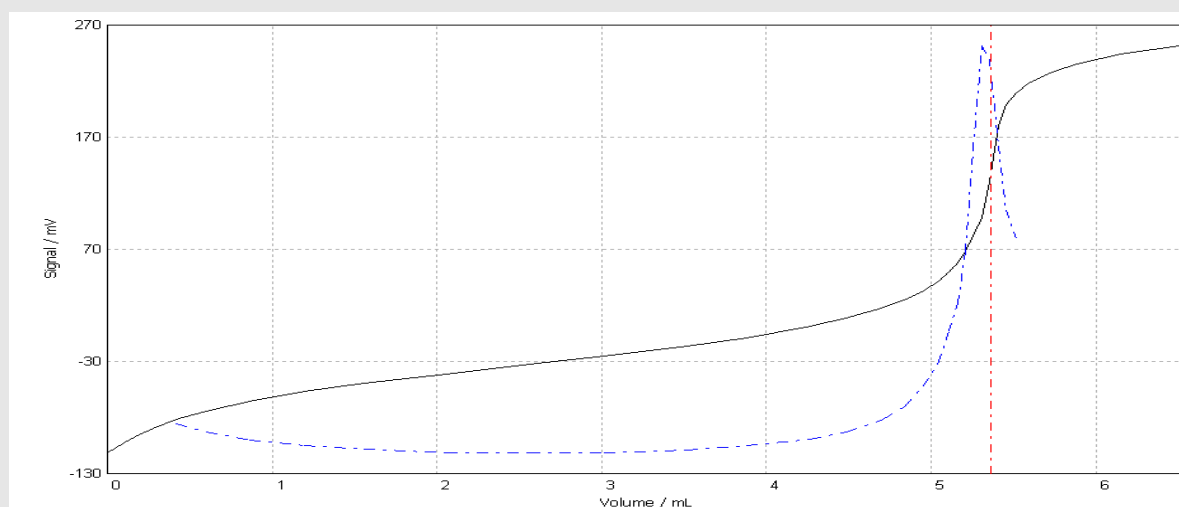


Table of measured values

	Volume mL	Increment mL	Signal mV	Change mV	1st deriv. mV/mL	Time s	Temperature °C
EQP1	0.0000	NaN	-111.5	NaN	NaN	0	25.0
	0.0500	0.0500	-107.4	4.1	NaN	3	25.0
	0.1000	0.0500	-102.5	4.9	NaN	6	25.0
	0.1680	0.0680	-96.8	5.7	NaN	9	25.0
	0.2765	0.1085	-89.3	7.5	NaN	12	25.0
	0.4180	0.1415	-81.5	7.8	52.68	15	25.0
	0.6010	0.1830	-74.0	7.5	42.23	18	25.0
	0.8665	0.2655	-65.7	8.3	32.40	21	25.0
	1.1945	0.3280	-57.0	8.7	24.96	24	25.0
	1.5585	0.3640	-49.7	7.3	20.04	27	25.0
	3.5155	0.5000	-16.1	9.4	19.93	40	25.0
	3.8780	0.3625	-9.0	7.1	24.71	43	25.0
	4.2650	0.3870	1.4	10.4	33.90	46	25.0
	4.4850	0.2200	8.4	7.0	43.10	49	25.0
	4.6960	0.2110	17.4	9.0	57.90	52	25.0
	4.8370	0.1410	24.9	7.5	75.32	55	25.0
	4.9570	0.1200	33.1	8.2	102.10	58	25.0
	5.0475	0.0905	41.7	8.6	133.89	61	25.0
	5.1085	0.0610	49.9	8.2	173.17	64	25.0
	5.1585	0.0500	58.2	8.3	209.77	67	25.0
	5.2085	0.0500	68.6	10.4	275.43	70	25.0
	5.2585	0.0500	82.1	13.5	405.88	74	25.0
	5.3085	0.0500	98.5	16.4	540.76	77	25.0
	5.3585	0.0500	130.8	32.3	521.68	80	25.0
	5.359609	NaN	131.9	NaN	551.97	NaN	NaN
	5.4085	0.0500	180.1	49.3	410.68	85	25.0
	5.4585	0.0500	198.8	18.7	328.75	88	25.0
	5.5150	0.0565	209.7	10.9	292.63	91	25.0
	5.5950	0.0800	218.3	8.6	NaN	94	25.0
	5.7295	0.1345	228.1	9.8	NaN	97	25.0
	5.8885	0.1590	235.7	7.6	NaN	100	25.0
	6.1485	0.2600	243.8	8.1	NaN	104	25.0
	6.5435	0.3950	252.0	8.2	NaN	106	25.0

Comments

- Rondo sample changer with membrane pump was used. The conditioning time was set to 30 s along with rinsing was defined to 10 mL. In this way, the electrode is cleaned with deionized water before starting the subsequent sample.
- The mean value of the titer is automatically stored as part of the setup by the function TITER.
- The purity of the primary standard, TRIS, is $p = 100.00\%$

Literature:

Mettler-Toledo Application M011 (V1.0)

Method

001 Title

Type	Titer determination
Compatible with	T50 / T70 / T90
ID	TiterH2SO4
Title	Titer 0.1 mol/L ½ H2SO4
Author	admin
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002 Sample (Titer)

Titrant	½ H2SO4
Concentration	0.1 mol/L
Standard	TRIS
Type of standard	Solid
Entry type	Weight
Lower limit	0.05 g
Upper limit	0.10 g
Density	1.0 g/mL
Correction factor	1.0
Temperature	25.0°C
Entry	Arbitrary

003 Titration stand (Rondo/Tower A)

Type	Rondo/Tower A
Titration stand	Rondo60/1A
Lid Handling	No

004 Pump

Auxiliary reagent	Water
Volume	50 mL
Condition	No

005 Stir

Speed	30 %
Duration	120 s
Condition	No

006 Titration (EQP) [1]

Titrant

Titrant	½ H2SO4
Concentration	0.1 mol/L

Sensor

Type	pH
Sensor	DG111-SC
Unit	mV

Temperature acquisition

Temperature measurement	No
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Stir

Speed	30%
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Predispense

Mode	None
Wait time	0 s

Control

Control	User
Titrant addition	Dynamic
dE (set value)	8.0 mV
dV (min)	0.05 mL
dV (max)	0.5 mL
Mode	Equilibrium controlled
dE	1.0 mV
dt	1 s
t (min)	3 s
t (max)	30 s

Evaluation and recognition

Procedure	Standard
Threshold	200.0 mV/mL
Tendency	Positive
Ranges	0
Add. EQP criteria	No

Termination

At Vmax	10.0 mL
At potential	No
At slope	No
After number of recognized EQPs	Yes
Number of EQPs	1
Combined termination	
Criteria	No

Accompanying stating

Accompanying stating	No
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Condition

Condition	No
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007 Calculation R1

Result	Titer
Result unit	--
Formula	$R1 = m / (VEQ \cdot c \cdot C)$
Constant	$C = M / (10 \cdot p \cdot z)$
M	M[TRIS]
z	z[TRIS]
Decimal places	5
Result limits	No
Record statistics	Yes
Extra statistical func.	No
Send to buffer	No
Condition	No

008 Rinse

Auxiliary reagent	Water
Rinse cycles	1
Vol.per cycle	10mL
Position	Current Position
Drain	No
Condition	No

009 Conditioning

Type	Fix
Interval	1
Position	Conditioning beaker
Time	30s
Speed	60%
Condition	No

010 End of sample

011 Titer

Titrant	½ H2SO4
Concentration	0.1 mol/L
TITER	Mean[R1]
Limits	No
Condition	No

012 Calculation R2

Result	Mean Titer
Result unit	--
Formula	$R2 = \text{Mean}[R1]$
Constant	C=1
M	M[TRIS]
Z	z[TRIS]
Decimal places	5
Result limits	No
Record statistics	Yes
Send to buffer	No
Condition	No